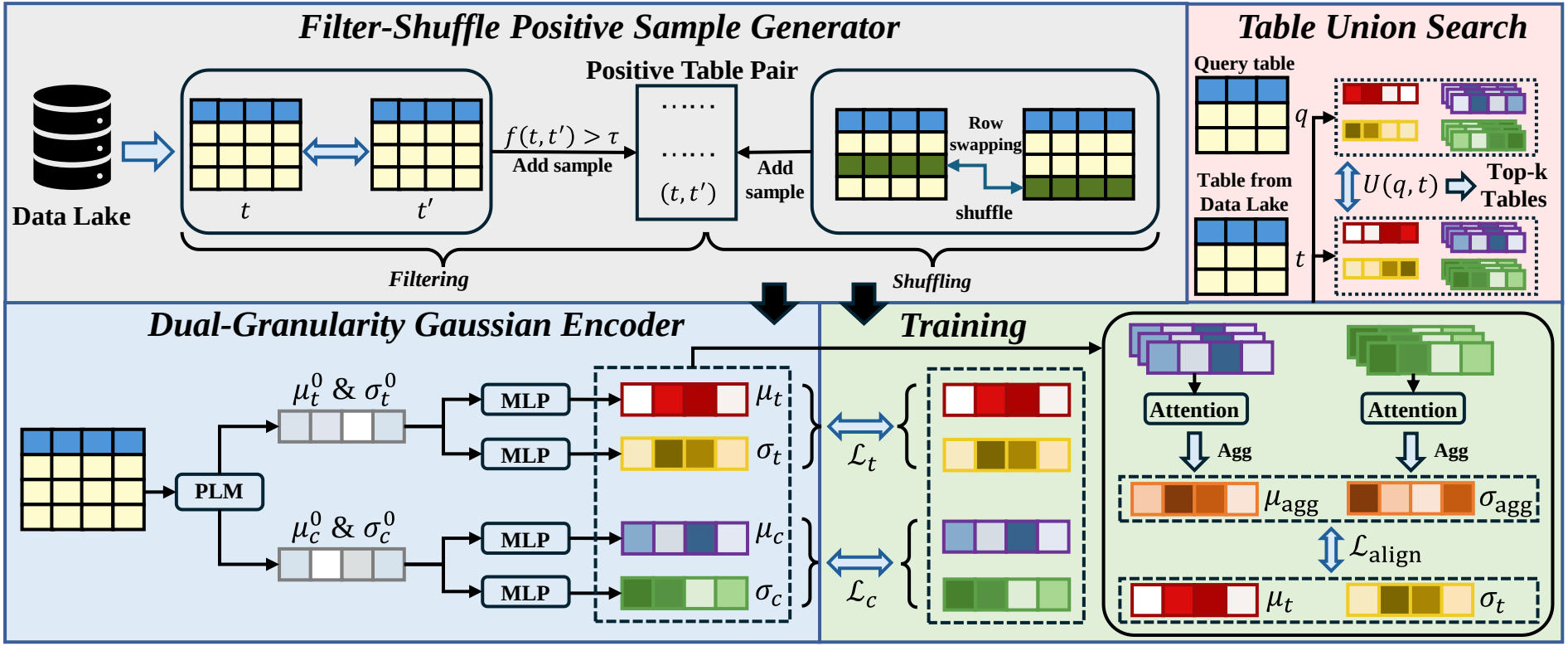

Filter-Shuffle Positive Sample Generator
Data Lake
t
t′
f(t, t′) > τ
Add sample
Positive Table Pair
(t, t′)
Add sample
Row swapping
shuffle
Filtering
Shuffling
Table Union Search
Query table
q
Table from Data Lake
t
U(q, t)
Top-k Tables
Dual-Granularity Gaussian Encoder
PLM
μ_t^0 & σ_t^0
μ_c^0 & σ_c^0
MLP
MLP
MLP
MLP
μ_t
σ_t
μ_c
σ_c
Training
ℒ_t
ℒ_c
Attention
Attention
Agg
Agg
μ_agg
σ_agg
ℒ_align
μ_t
σ_t